

Course “Theoretical Ecotoxicology: Concepts and applications”

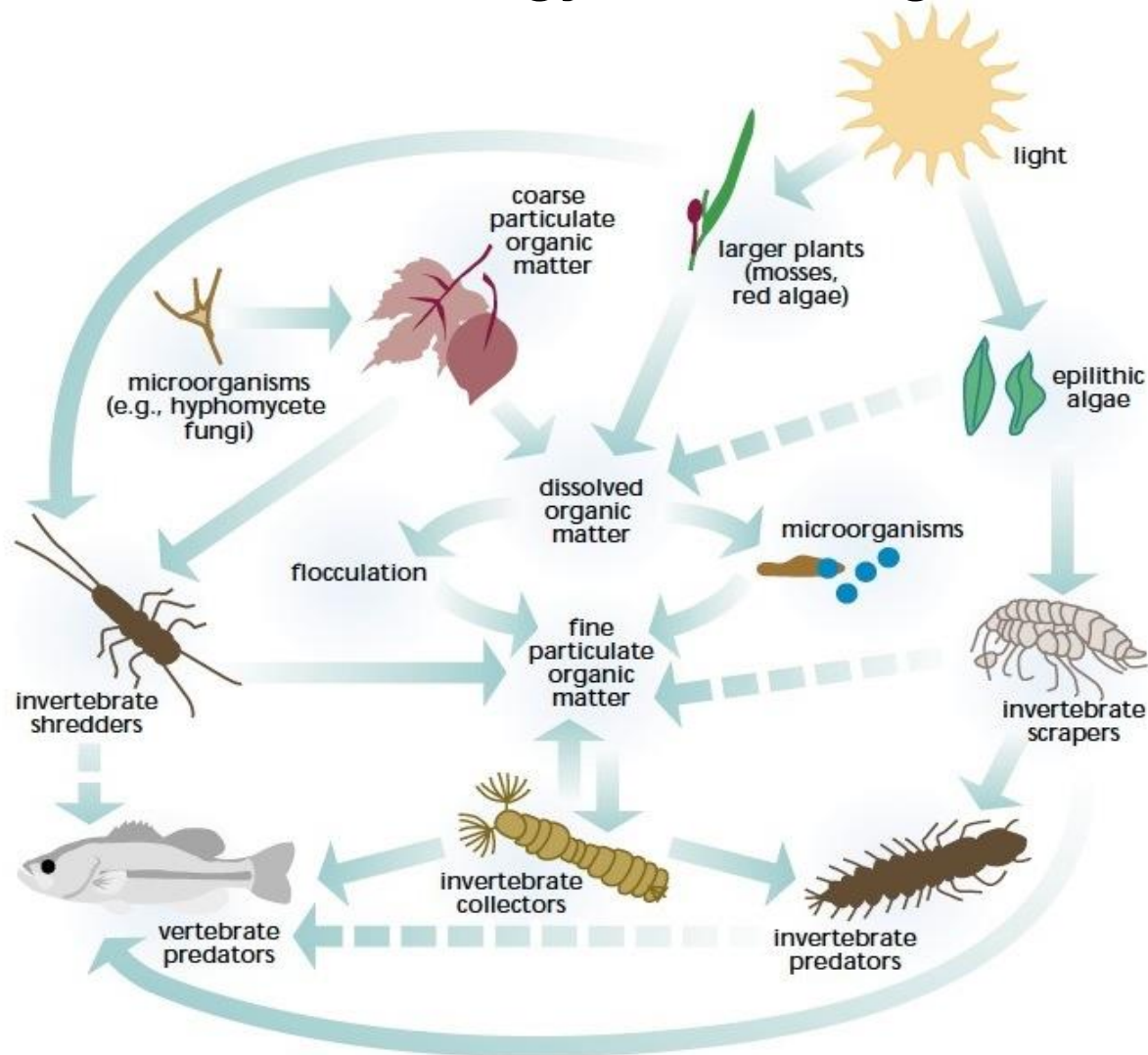
Introduction to Ecological Modelling I

Hanh H. Nguyen

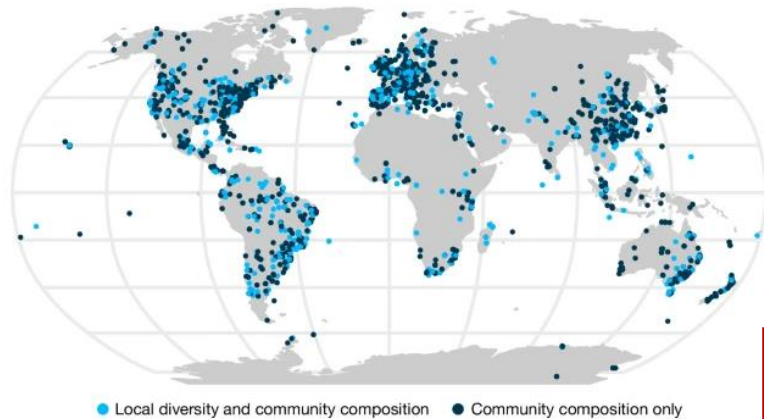
University of Duisburg-Essen
& Research Center One Health Ruhr

Introduction

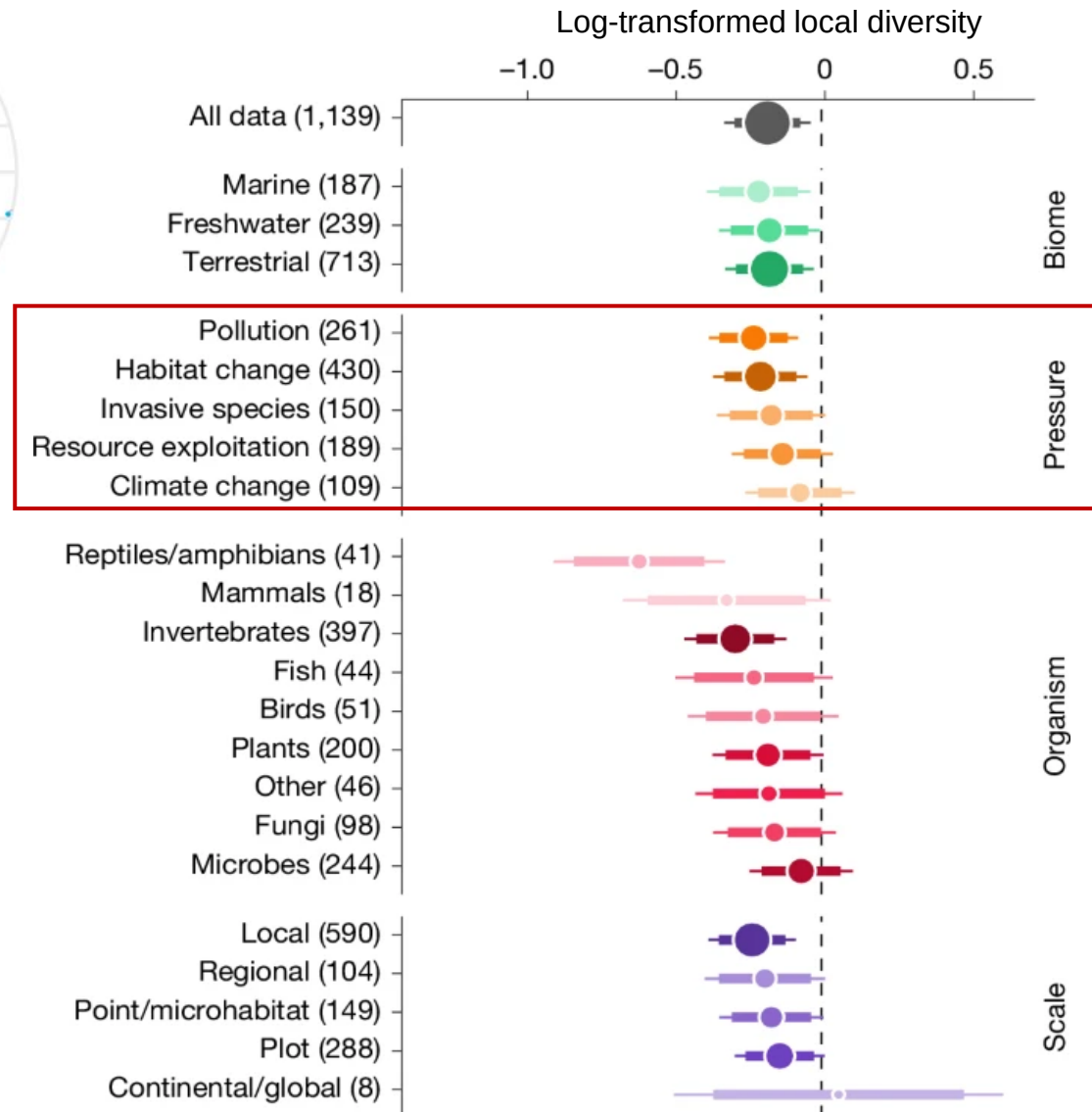
Why interested in **ecology** and **ecological modelling**?



The global human impact on biodiversity

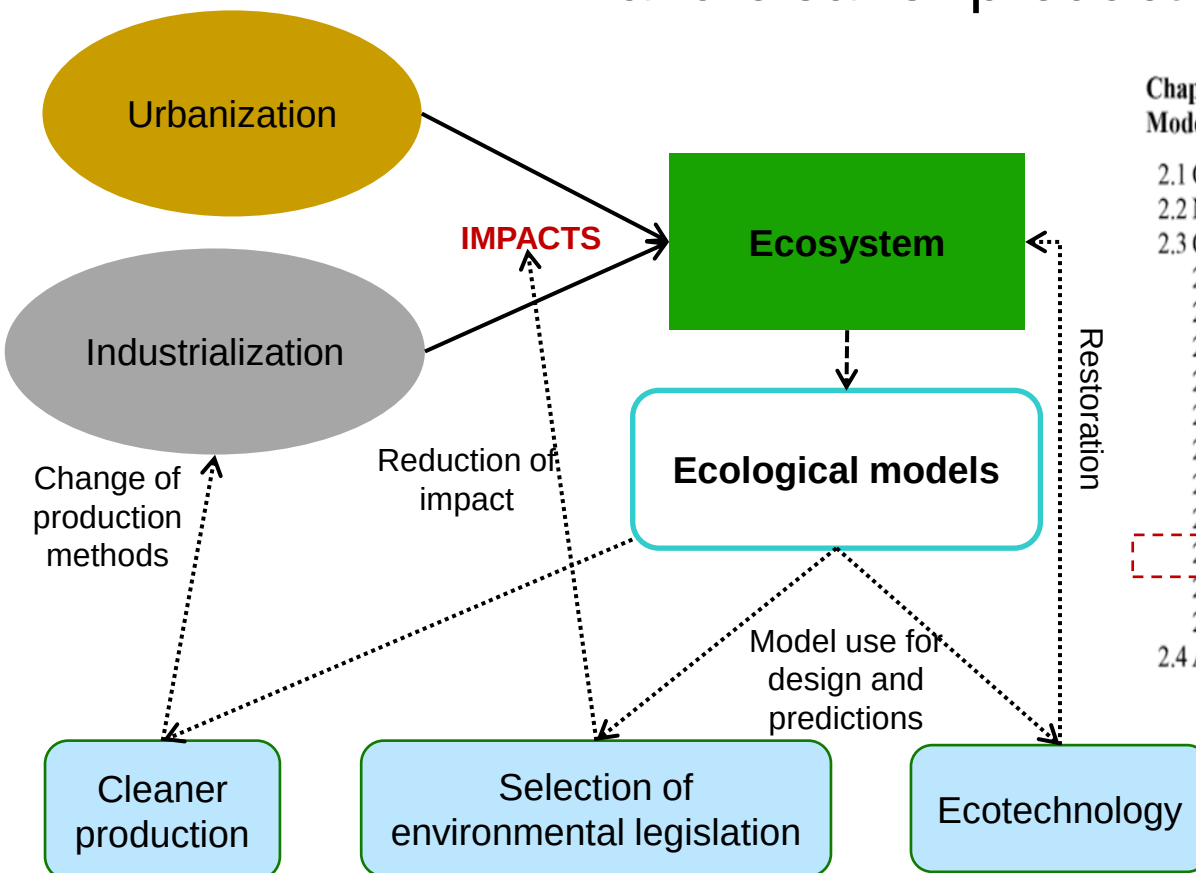


Biodiversity crisis caused by multiple anthropogenic stressors, including pesticides



Ecological Modelling

Application of ecological models to select environmental technology, environmental legislation, and cleaner production



Chapter 2

Model types

2.1 Classification of models.....

2.2 Model types – an overview

2.3 Characteristics of the model types

2.3.1 Dynamic biogeochemical and bioenergetic models.....

2.3.2 Steady state biogeochemical models.....

2.3.3 Population dynamic models.....

2.3.4 Structurally dynamic models

2.3.5 Fuzzy models

2.3.6 Artificial neural networks

2.3.7 Spatial models.....

2.3.8 Individual based models and cellular automata

2.3.9 Ecotoxicological models.....

2.3.10 Stochastic models

2.3.11 Hybrid models

2.4 Applicability of the different model types.

Learning topics

I. Non-spatial approaches - Population dynamics:

- Population growth model
- Species interaction models

II. Spatially-explicit approaches - Spatio-temporal dynamics:

- Population in space
- Metapopulation models

Robust models to realistically reflect natural systems:

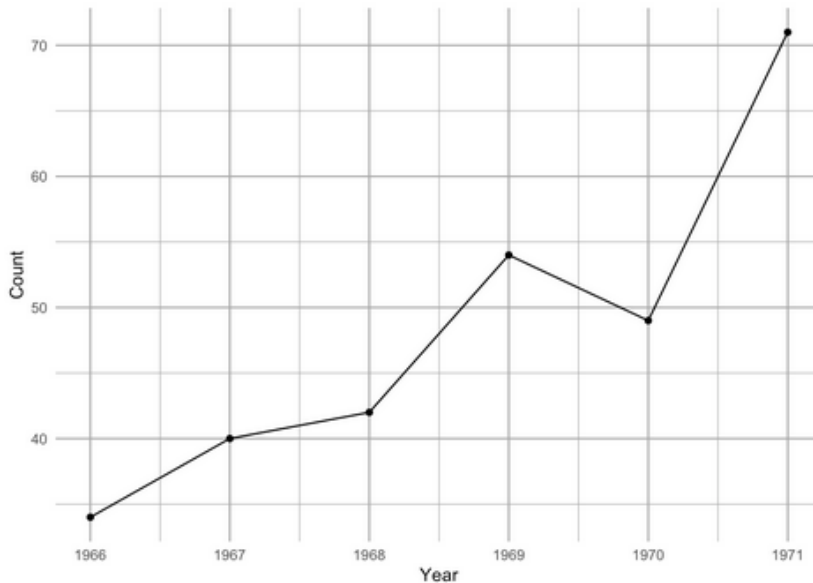
- (i) use empirical data for sound hypothesis testing
- (ii) uncertainty reduction

(Arhonditsis et al. 2025 – “Ecological models for tomorrow’s solutions”)

Population growth models

Density-independent population growth

Song sparrow (*Melospiza melodia*) counts
($\Sigma 38$ years) in Darrrtown (Ohio, USA)



$$\frac{dN}{dt} = rN$$

- N = population
- r = growth rate

Analytical solution:

$$N_t = N_0 * e^{rt}$$



<https://ebird.org/species/sonspa>

- A population grows exponentially
- Main assumptions:
 - All individuals are identical
 - Unlimited resources
 - Constant per capita growth rate

Exercise 1

Density-independent population growth

- 1) Run the density-independent population growth model in R (find and fix errors in the code)
- 2) Manipulate r and N and see how the model reacts.

Does N affect model result pattern?

```
# Load library
library(deSolve)

# Model function - Exponential growth  $dN/dt = r*N$ 
dens_indep <- function(times, init, parms){
  with(as.list(c(init,parms)), {
    dN_dt = r
    return(list(c(dN_dt)))
  })
}

# Model parameters
times <- seq(from = 0, to = 120, by = 1)
initial <- c(N = 10)
parms <- c(r = 0.03)

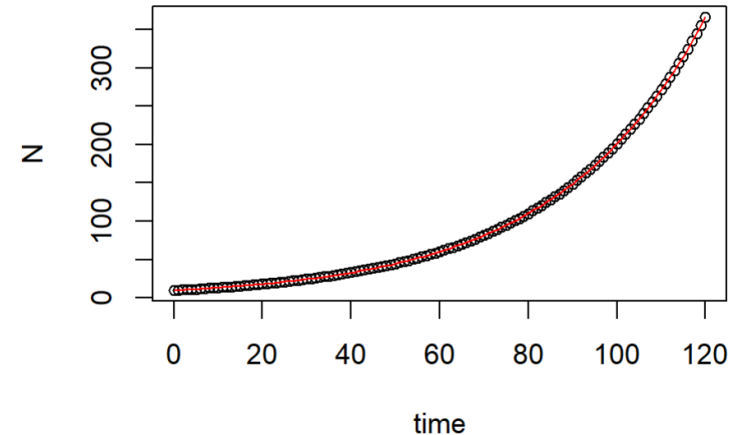
# Model simulation run using ODE
result_dens_indep <- ode(func = dens_indep, times = times, y = init)
head(result_dens_indep)

# Model plot
plot(N ~ time, data = result_dens_indep, pch = 1, cex = 0.8, main = "Density-independent growth model")

# Optional- Analytical solution:  $N(t) = N_0 * e^{rt}$ 
curve(init[1]*exp(parms[1]*x), col = "red", lwd = 2, add = T)

## Questions:
# 1. Run the model in R - resolve errors in model function, model parameters, and model simulation
# 2. Test if changing population density  $N$  (e.g.,  $N = 2$  or  $200$ ) and growth rate  $r$  affect the result pattern
```

Density-independent growth model



Exercise 1

Density-independent population growth

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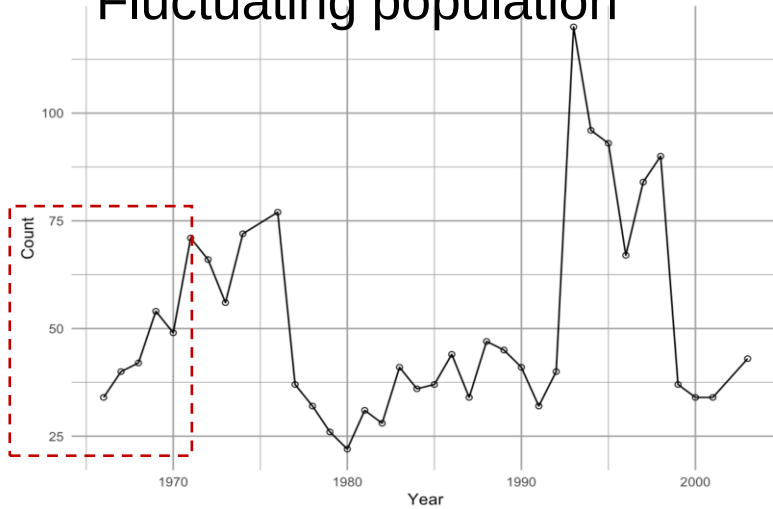
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```

Density-dependent population growth

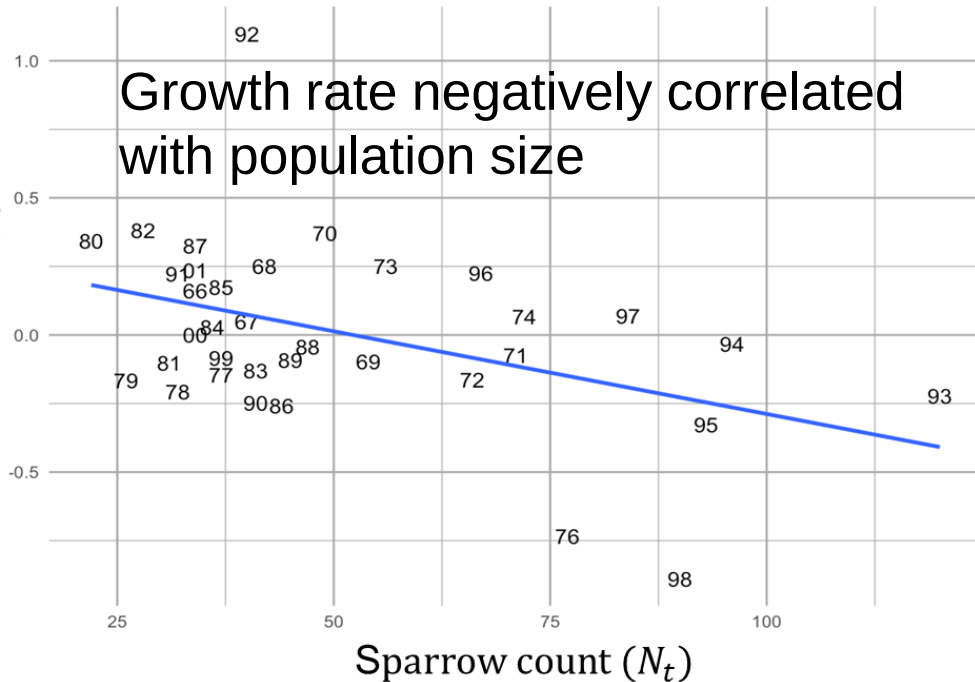
Song sparrow (*Melospiza melodia*) counts
($\Sigma 38$ years) in Darrrtown (Ohio, USA)

Fluctuating population



$$\text{Growth rate } r_t = \frac{N_{t+1}}{N_t}$$

Growth rate (r_t)



<https://ebird.org/species/sonspa>

<https://hankstevens.github.io/Primer-of-Ecology/expo.html>

Density-dependent population growth

Assumption:

Increased density causes intraspecific competition due to limited resources, resulting in linear $\uparrow$ **death** and $\downarrow$ **in births**

Continuous logistic model (Verhulst, 1838; Verhulst, 1845):

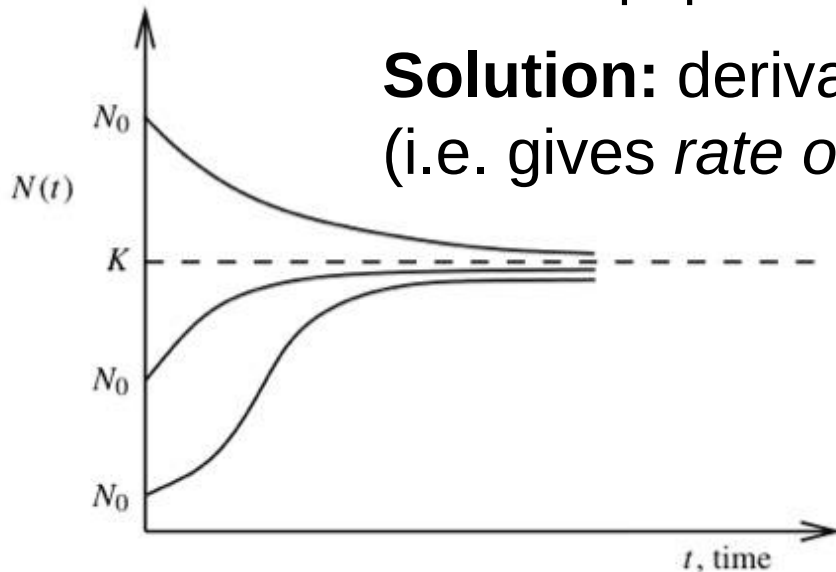
$$\frac{dN}{dt} = rN \left[1 - \frac{N}{K} \right]$$

- N = population
- r = intrinsic (per capita) growth rate
- $K = \frac{1}{\alpha}$: carrying capacity, i.e. maximum population size that environment can support

Solution: derivative from the differential equation (i.e. gives *rate of change* over time)

Analytical solution:

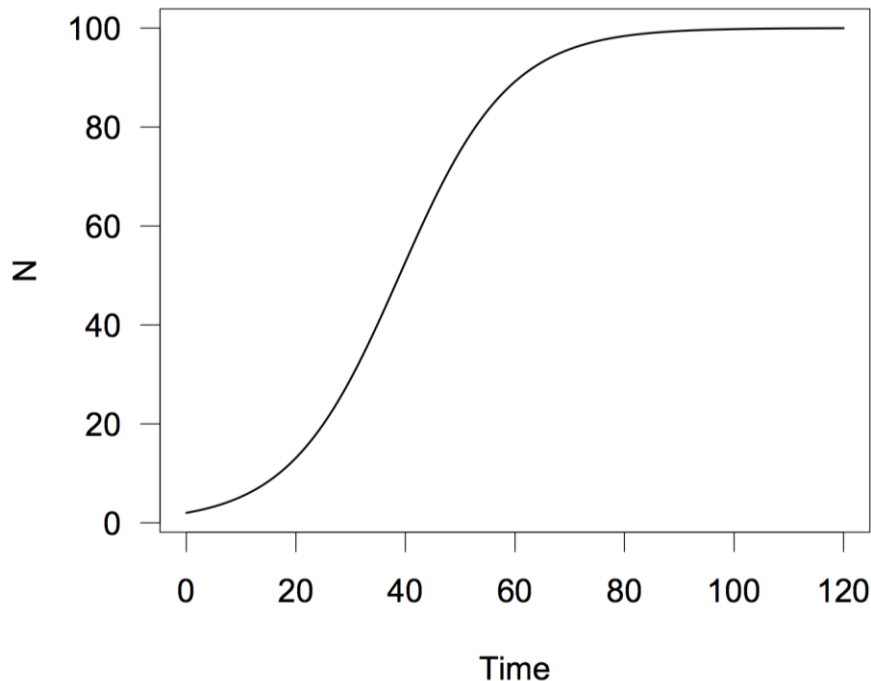
$$N(t) = \frac{K}{1 + \left(\frac{K - N(0)}{N(0)} \right) e^{-rt}}$$



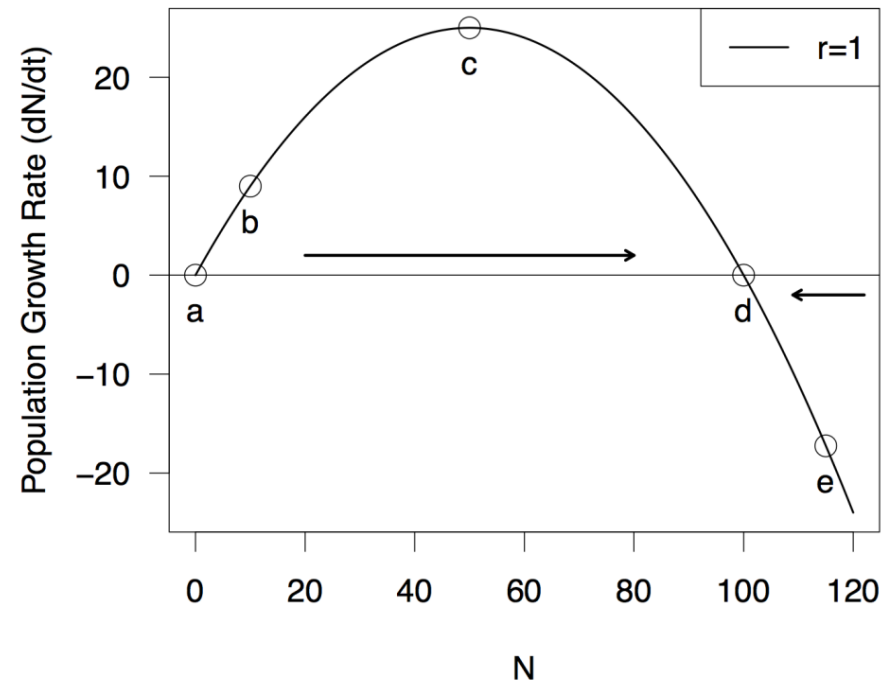
Stability at equilibria

$$\frac{dN}{dt} = rN(1 - \alpha N)$$

⇒ stability at equilibria: $\frac{dN}{dt} = 0$



a = if perturbed goes towards 100
b = positive growth rate
c = max. positive growth rate
d = equilibrium (attractor)
e = negative growth



Stability at equilibria

Question: Which N yields **maximum growth rate**?

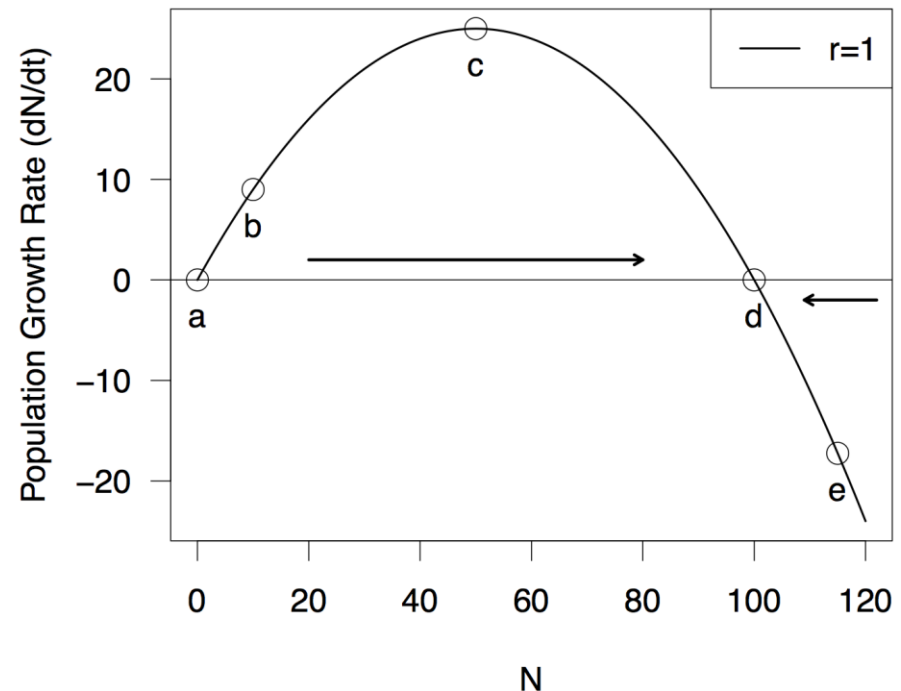
Solution: Derive dN/dt for $N \rightarrow$ **Partial derivative** (function of N)

We define:

$$\begin{aligned}\frac{dN}{dt} = \dot{N} &\Leftrightarrow \frac{\partial \dot{N}}{\partial N} = \frac{\partial rN(1 - \alpha N)}{\partial N} = \\ &= r(1 - \alpha N) + rN(-\alpha) = r(1 - 2\alpha N)\end{aligned}$$

Maximum growth:

$$\begin{aligned}\frac{\partial \dot{N}}{\partial N} = 0 &\Rightarrow r(1 - 2\alpha N) = 0 \\ \Leftrightarrow N &= \frac{1}{2\alpha} = \frac{K}{2}\end{aligned}$$



Stability at equilibria

Question: What is return rate for small perturbations from d?

We define:

At a: $N = 0 \Leftrightarrow \frac{\partial \dot{N}}{\partial N} = r(1 - 2\alpha N) = r(1 - 2\alpha * 0) \Leftrightarrow r = \frac{\partial \dot{N}}{\partial N}$

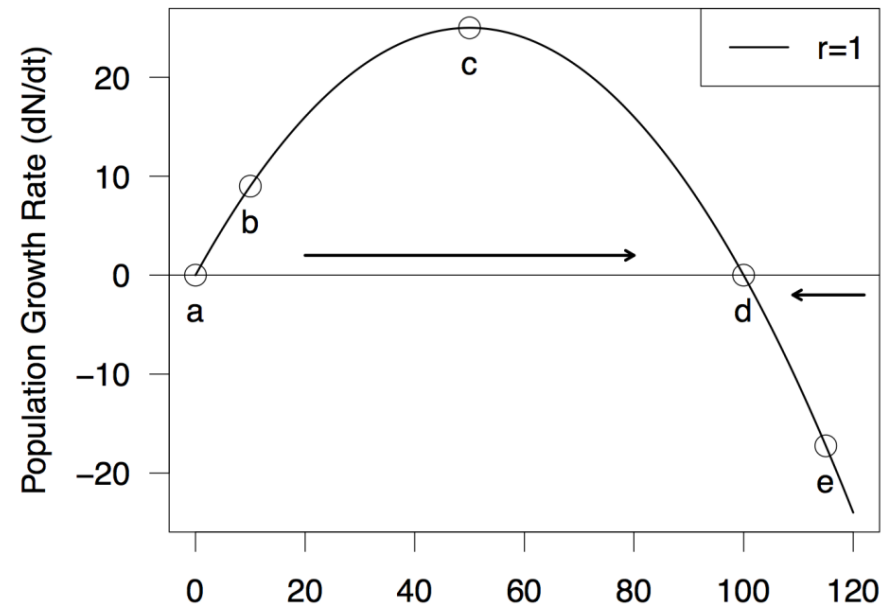
At d: $N = K = \frac{1}{\alpha} \Leftrightarrow \frac{\partial \dot{N}}{\partial N} = r(1 - 2\alpha N) = r\left(1 - 2\alpha \frac{1}{\alpha}\right) \Leftrightarrow -r = \frac{\partial \dot{N}}{\partial N}$

The return rate at $N = 0$ is r , which is positive if $r > 0$.

→ **unstable equilibrium (trivial).**

A negative value indicates that perturbation from equilibrium will decline through time

→ **stable equilibrium (attractor).**



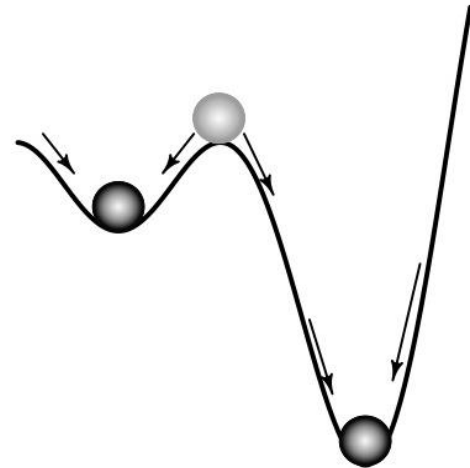
Exercise 2

Density-dependent population growth

- 1) Run the density-dependent population growth model in R
- 2) Manipulate r and α and see how the model reacts.
- 3) For which values of N is the model in equilibrium?

$$\frac{dN}{dt} = rN(1 - \alpha N) \quad \Rightarrow \quad \frac{dN}{dt} = 0$$

- 4) Which equilibrium is stable?



Exercise 2

Solution to the first question:

1) Run the density-dependent population growth model

```
# Load library
library(simecol) # Simulation of Ecological Dynamic Systems

## Q1. Run the density-dependent population growth model
# Model using differential equation ODE
dens_dep <- new("odeModel",      # model class (odeModel, gridModel, rwalkModel, indbasedModel, etc.)
  main = function(times, init, parms) {
    with(as.list(c(init, parms)), {
      dN_dt <- r * N * (1-alpha*N)
      list(c(dN_dt))
    })
  },
  parms = c(r = 0.3, alpha = 0.005),
  times = c(from = 0, to = 120, by = 1),
  init = c(N=2),
  solver= "lsoda" # numerical algorithm
)

# Run simulation
result_ddep <- sim(dens_dep)

# Model results
print(result_ddep, all=TRUE) # print all results
head(result_ddep@out)       # show only results from output matrix 'out'

# Model plot
plot(result_ddep@out[ , 1], result_ddep@out[ , 2], type = "l", xlab = "Time", ylab = "Count (N)",
  las = 1, lwd = 2, col = "blue")
```


Limitations of approach

- No species interactions included, which may prolong or hinder recovery
- Spatial dynamics (e.g. dispersal) not considered
- A simple logistic growth (i.e. a single population stage, no life or spatial stage separation)

Note: Some insects have more complex life-cycles (e.g., Merolimnic insects have aquatic or semi-aquatic egg and larval stages but emerge and reproduce on land)

Other relevant terms and topics

- **Theta-Logistic model:** extension of logistic population model, for non-linear density effects when approaching K. Theta controls shape of the density dependence: $\frac{dN}{dt} = rN \left(1 - \left(\frac{N}{K} \right)^\theta \right)$
- **Allee effect model:** per capita growth rate increases with population size, while sparse populations below critical threshold density (A) face extinction (e.g., due to reduced mate availability or predator defense challenge).

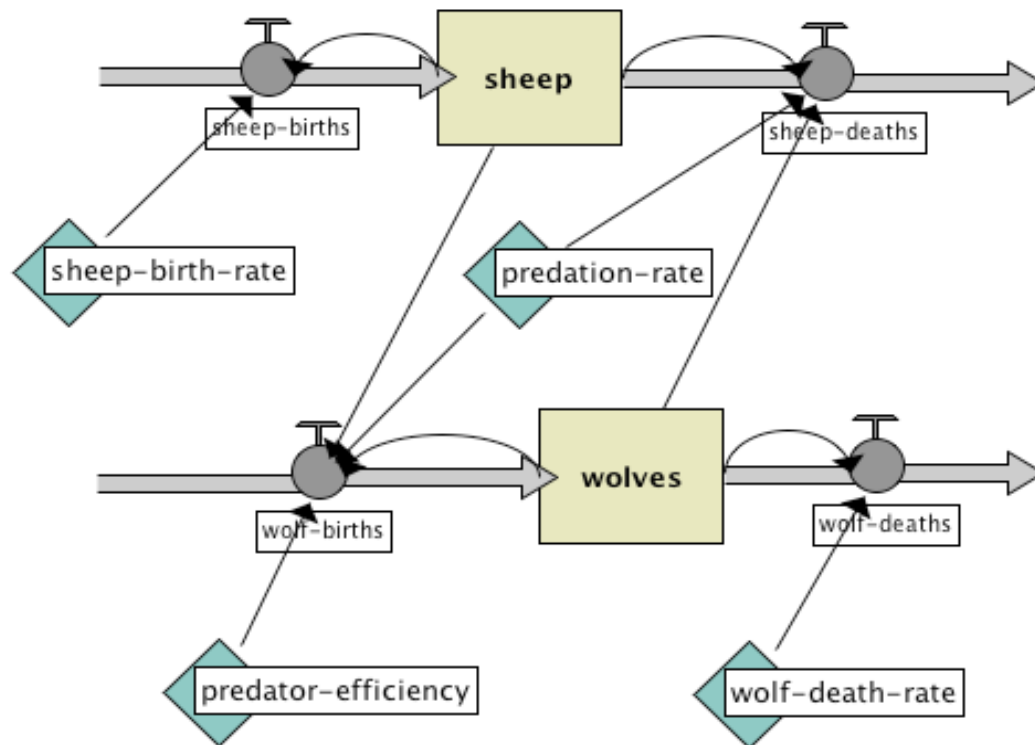
$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right) \left(\frac{N}{A} - 1 \right)$$

- **Discrete-time model:** track population size at distinct time intervals (e.g., annual breeding seasons). λ is per capita growth rate: $N_{t+1} = \lambda N_t$

Species interaction models

Species Interactions

- Classical Lotka-Volterra model:
 - **competitive interactions** between two species
 - species **predator-prey interactions**



(Lotka-Volterra, 1920)

Species Interactions: Competition

- Lotka-Volterra model for competition

$$\frac{dN_1}{dt} = r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1} \right)$$

N = population

r = growth rate

t = time

$$\frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2 + \alpha_{21} N_1}{K_2} \right)$$

K = carrying capacity

α = interspecific competition (between different species)

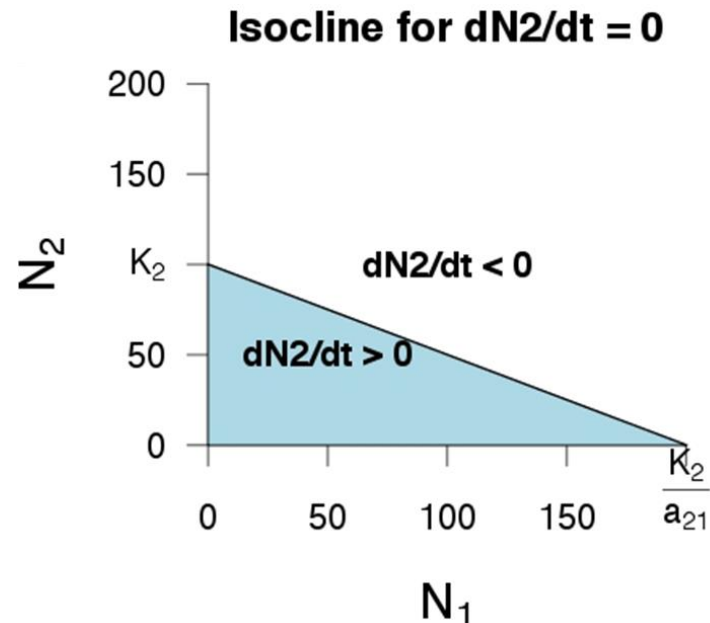
- α_{12} : effect of species 2 on species 1
- α_{21} : effect of species 1 on species 2

What does an α of 0.5 and 2 mean?

0.5 and 2 means that one individual of the other species has half and double the influence on the growth rate as one individual from the same species, respectively

Zero-growth isoclines for competition models

- **Zero isoclines:** the lines where rate of change (dN/dt) of each state variable is zero, as the function of their values (N)
- **Phase plane:** a X-Y graph with values of state variables N_1 and N_2 plotted on X and Y-axis
- For a given species, phase space shows growth/decline areas



Zero-growth isoclines for competition models

- **Equilibrium point:** Intersection of isoclines (i.e., rate of change of the two state variables are simultaneously 0)

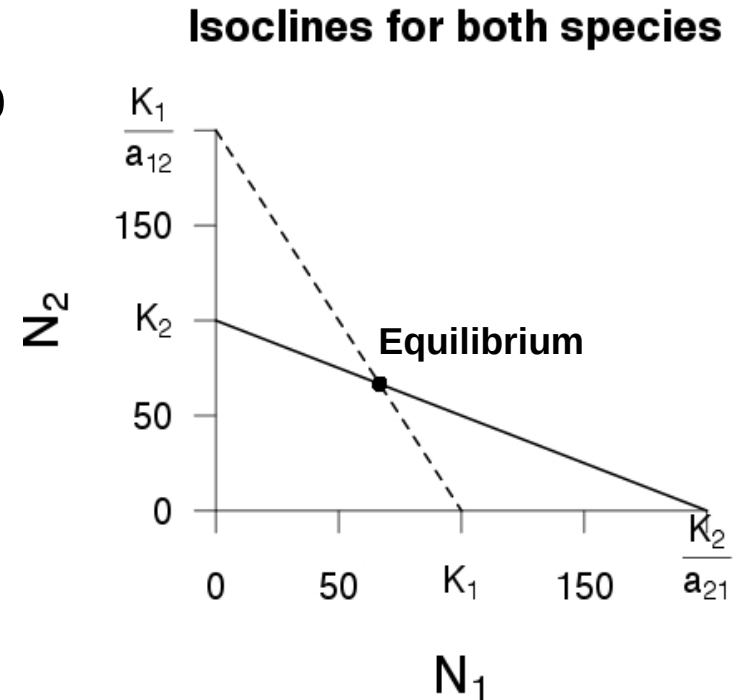
- Isocline at equilibrium: $\frac{dN_1}{dt} = 0$

$$\Rightarrow r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1} \right) = 0$$

$$\Leftrightarrow 1 - \frac{N_1 + \alpha_{12} N_2}{K_1} = 0$$

$$\Leftrightarrow N_1 = K_1 - \alpha_{12} N_2$$

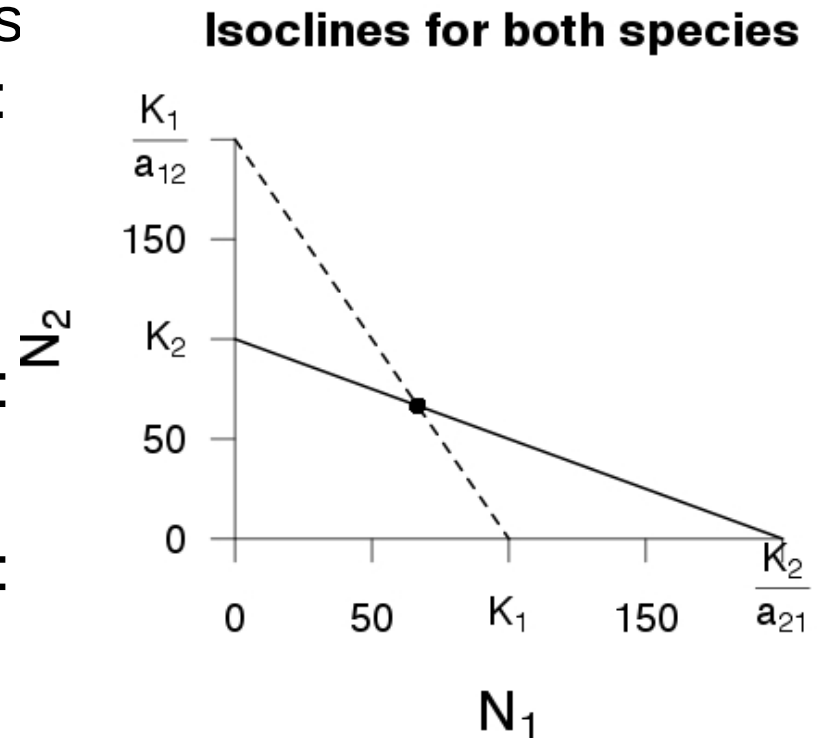
- Isocline for species 1 equilibrium: $K_1 = N_1 + \alpha_{12} N_2$
Equilibrium: $N_1 = 0$; a line at $N_1 = K_1$ and $N_2 = K_1 / \alpha_{12}$
- Isocline for species 2 equilibrium: $K_2 = N_2 + \alpha_{21} N_1$
Equilibrium: $N_2 = 0$; a line at $N_2 = K_2$ and $N_1 = K_2 / \alpha_{21}$



Analytical analysis of equilibria

Important insights:

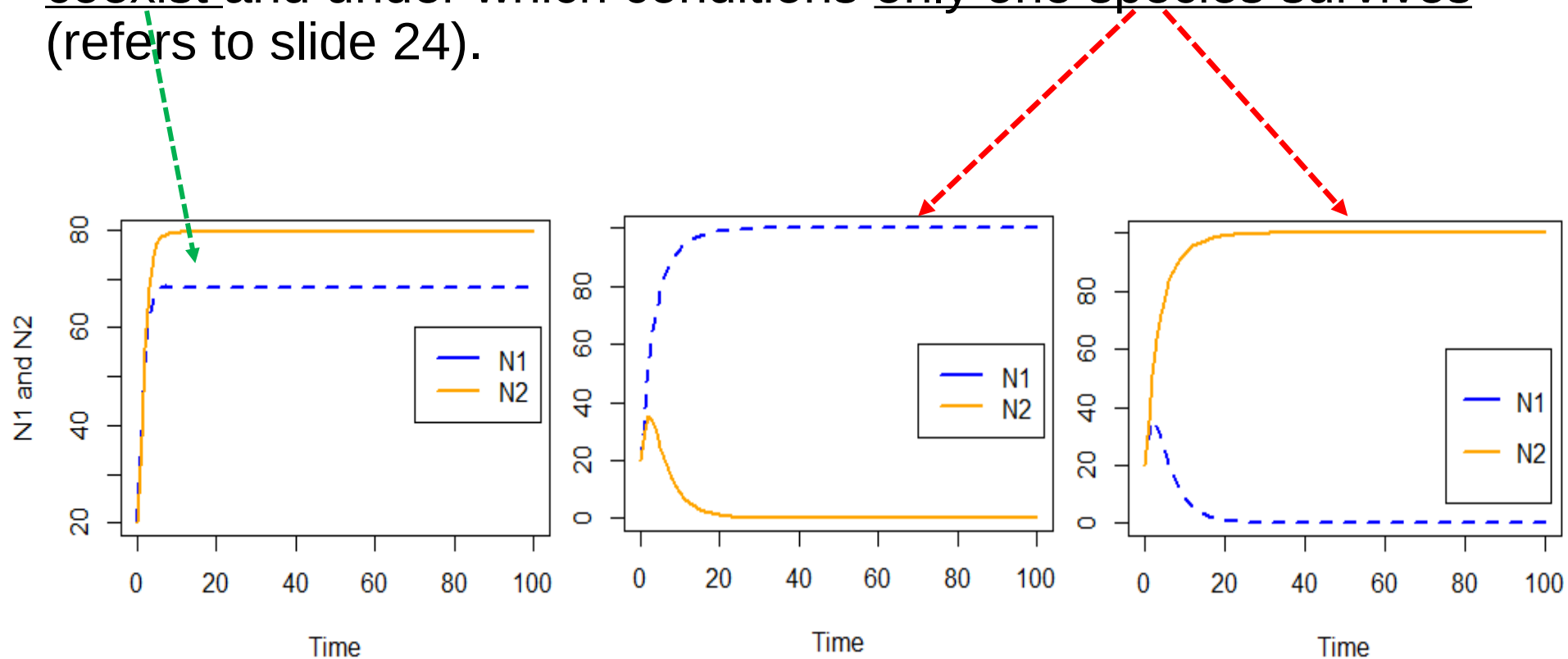
- Equilibrium independent of growth rate r
- Equilibrium depends only on competition coefficients α
- **Coexistence:** each species limits itself more than it limits the other:
 $K_1 / \alpha_{12} > K_2$ and $K_2 / \alpha_{21} > K_1$
- **Only one species survive:**
 - Species 1 outcompete species 2:
 $K_1 / \alpha_{12} > K_2$ and $K_2 / \alpha_{21} < K_1$
 - Species 2 outcompete species 1:
 $K_1 / \alpha_{12} < K_2$ and $K_2 / \alpha_{21} > K_1$



Exercise 3

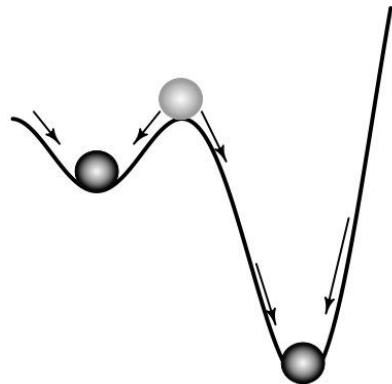
1) Implement the species competition model in R

2) Try to find out for which values of K and α the species can coexist and under which conditions only one species survives (refers to slide 24).

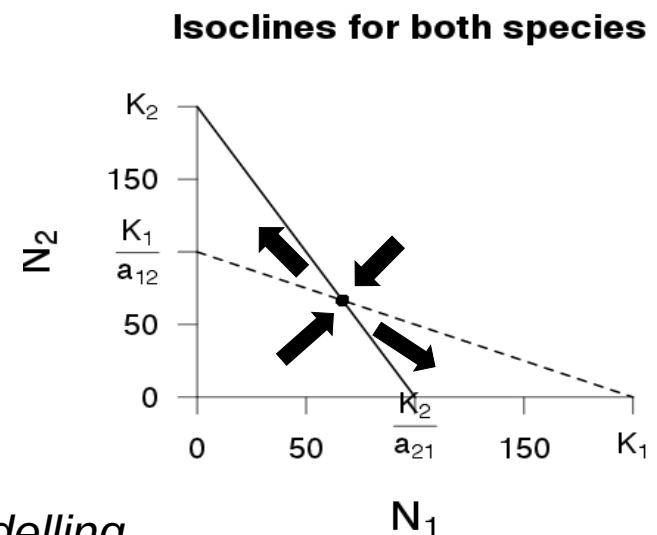
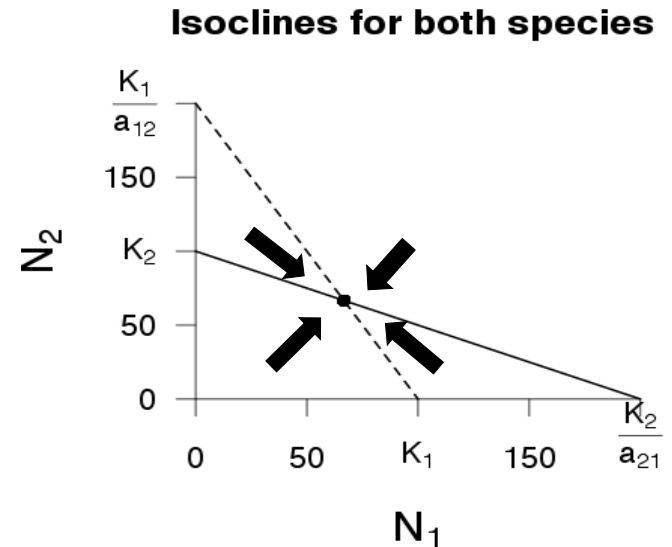


Stability of Equilibrium for competition

- Arrows indicate how N changes in vicinity of equilibrium points
- Three cases of stability:
 - Arrows point to the equilibrium: **stable**
 - Arrows diverge from the equilibrium: **unstable**
 - Arrows point to the equilibrium on one side and point away from the point on the other side: **unstable (saddle) point**



Soetaert and Herman 2009 – A practical guide to Ecological modelling



Analytical analysis of equilibria

- Beside trivial equilibria (N_1 or $N_2 = 0$), the equilibrium is a point belonging to both isoclines:
$$\begin{aligned}N_1 &= K_1 - \alpha_{12}N_2 \\N_2 &= K_2 - \alpha_{21}N_1\end{aligned}$$
- Substitute N_2 in equation for N_1 :

$$\begin{aligned}N_1 &= K_1 - \alpha_{12}(K_2 - \alpha_{21}N_1) \\ \Leftrightarrow N_1 &= K_1 - \alpha_{12}K_2 + \alpha_{12}\alpha_{21}N_1 \\ \Leftrightarrow N_1(1 - \alpha_{12}\alpha_{21}) &= K_1 - \alpha_{12}K_2 \\ \Leftrightarrow N_1 &= \frac{K_1 - \alpha_{12}K_2}{1 - \alpha_{12}\alpha_{21}}\end{aligned}$$

=> Repeat the analysis for N_2 :

$$N_2 = \frac{K_2 - \alpha_{21}K_1}{1 - \alpha_{12}\alpha_{21}}$$

Important insights:

- equilibrium of N_1 is independent of r
- for $\alpha_{12} = 0$, species N_1 reaches its own K
- effect of α_{12} on N_1 can be balanced by α_{21}

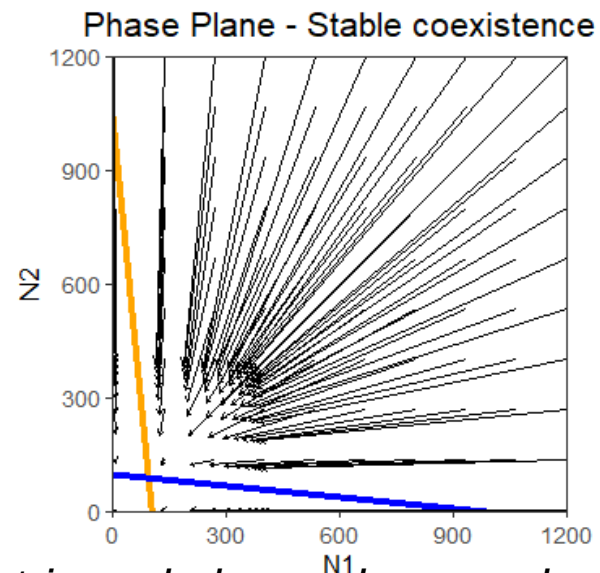
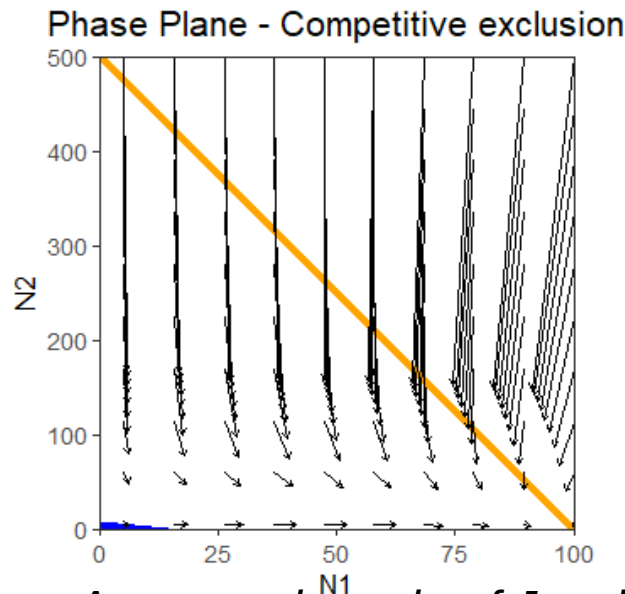
Stability analysis with Jacobian matrix

- Jacobian matrix for the model is given as:

$$\begin{pmatrix} \frac{\dot{N}_1}{\partial N_1} & \frac{\dot{N}_1}{\partial N_2} \\ \frac{\dot{N}_2}{\partial N_1} & \frac{\dot{N}_2}{\partial N_2} \end{pmatrix} = \begin{pmatrix} r_1 - 2r_1\alpha_{11}N_1 - r_1\alpha'_{12}N_2 & -r_1\alpha'_{12}N_1 \\ -r_2\alpha_{21}N_2 & r_2 - 2r_2\alpha_{22}N_2 - r_2\alpha_{21}N_1 \end{pmatrix}$$

Impact of changes of the population on itself

Impact of changes of the other population



An example code of Jacobian matrix and phase plane analyses
in the **Exercise_4_Additional.R**

Summary – Today's learning targets

- Population growth model:
 - Understand how **population dynamics** is implemented and explain influences of key parameters (r and α) on model results
 - Determine maximum growth rate and **stable equilibrium**
- Species interaction model:
 - Understand how **species interaction** is implemented in the model and the role of key parameters (K and α)
 - Under which conditions **species coexist (stable equilibrium)** and under which conditions do species tend to **extinction (unstable equilibrium)**